Employee Database Management System using MySQL

1. Project Objective

The **Employee Database Management System** is a relational database project developed using **MySQL** to efficiently store, organize, and manage employee-related information. including employee details, salaries, departments, designations, and locations. The project aims to apply SQL concepts such as **DDL commands, constraints, filtering, sorting, aggregation, grouping, joins, and window functions** to generate meaningful insights and support effective employee and workforce management.

2. Problem Statement

- **Develop** a structured database to store and manage employee information efficiently.
- **Organize** employee details such as employee ID, name, gender, date of joining, designation, salary, department, and location.
- **Maintain** data accuracy and consistency by applying appropriate constraints such as Primary Key, Foreign Key, `NOT NULL`, `UNIQUE`, and `CHECK`.
- **Retrieve** employee records based on specific conditions such as salary, joining date, designation, and age.
- **Analyze** salary and employee distribution across different departments and locations using aggregate functions and grouping.
- **Identify** departments with fewer than 3 employees and locations based on employee demographics.
- **Compare** employees across departments and locations using different types of SQL joins.
- **Rank** employees based on salary overall and within individual departments using window functions.
- **Evaluate** departmental salary patterns through running-total salary analysis to support workforce and compensation decisions.

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3. Attribute (Column /Features) Details:

Table 1: Departments

Attribute Name	Data Type	Description
Department_id	Int (Primary Key)	Unique identifier assigned to each department
Department_ name	Varchar (100)	Name of the Department

Table 2: Locations

Attribute Name	Data Type	Description
Location_id	Int (Primary Key)	Unique identifier assigned to each Location
Location_ name	Varchar (30)	Name of the Location

Table 3: Employees

Attribute Name	Data Type	Description
Employee_id	Int (Primary Key)	Unique identifier assigned to each Employees
Employee_ name	Varchar (30)	Name of the Location
Gender	Enum (M,F)	Employees's gender (Male or Female
Age	Int	Age of the employees in years
Hire-date	Date	Date of joining for Employees
Designation	Varchar (100)	Name of designation for Employees
Department_id	Int (Foreign Key)	Unique identifier assigned to each Department referenced to ForeignKey Constrint
Location_id	Int (Foreign Key)	Unique identifier assigned to each Locations referenced to ForeignKey Constrint
Salary	Decimal(10,2)	Employees Salary amount
Email	Varchar(100)	Email address for Employees

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4. Tools & Technologies

- **MySQL:** MySQL, SQL, DDL Commands, SQL Constraints, WHERE Clause, ORDER BY, GROUP BY, HAVING, Aggregate Functions, INNER JOIN, LEFT JOIN, RIGHT JOIN, CROSS JOIN, SELF JOIN, RANK(), DENSE_RANK(), Window Functions

5. Data Pre-Processing (MySQL)

Tasks Performed:

DDL Commands:

1. Database and Table Creation (CREATE):

Department table creation

```
4  create database if not exists employee_db;
5
6  use employee_db;
7
8                                          -- Create three tables with appropriate data types and relationships
9                                          -- 1. Department table creation
10
11 create table if not exists departments(
12 department_id int,
13 department_name varchar(100),
14
15 Primary Key (department_id)
16 );
17 desc departments;
18                                         -- 2. Location table creation
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
department_id	int	NO	PRI	NULL	
department_name	varchar(100)	YES		NULL	

Location table creation

```
18                                         -- 2. Location table creation
19
20 create table if not exists location(
21 location_id int primary key,
22 location_name varchar(30)
23 );
24 desc location;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
location_id	int	NO	PRI	NULL	
location_name	varchar(30)	YES		NULL	

Employees table creation

40

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(50)	YES		NULL	
gender	enum('M','F')	YES		NULL	
age	int	YES		NULL	
hire_date	date	YES		NULL	
designation	varchar(100)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	

Result 6

2. Table Alteration (ALTER)

- Add a new column named "email" to the Employees table to store employee email addresses.
- Modify the data type of the "designation" column in the Employees table to support a wider range of values.
- Drop the “age” column from the Employees table.
- Rename the “hire_date” column to “date_of_joining”

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(50)	YES		NULL	
gender	enum('M','F')	YES		NULL	
date_of_joining	date	YES		NULL	
designation	varchar(200)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	
email	varchar(100)	YES		NULL	

Result 7

3. Table Renaming (RENAME):

- Rename the "Departments" table to "Departments_Info"
- Rename the "Location" table to "Locations

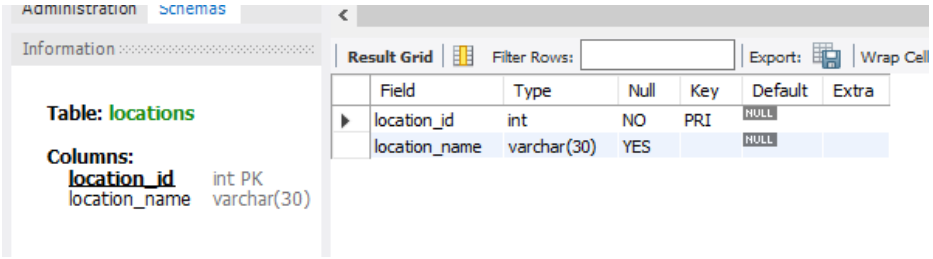
Information

Table: departments_info

Columns:
department_id int PK
department_name varchar(100)

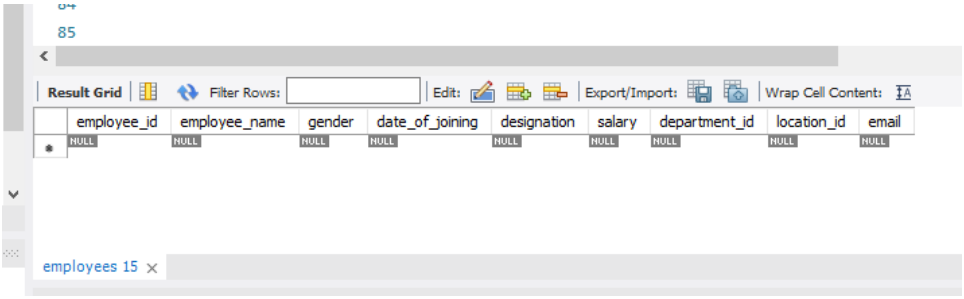
Result Grid | Filter Rows: | Export: | Wrap Cell Content

Field	Type	Null	Key	Default	Extra
department_id	int	NO	PRI	NULL	
department_name	varchar(100)	YES		NULL	



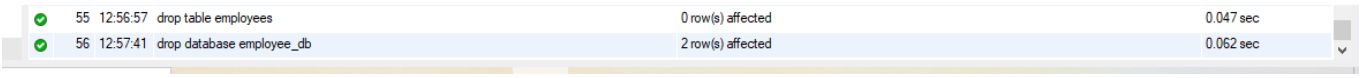
4. Table Truncation (TRUNCATE):

- Truncate the Employees table



5. Database & Table Dropping (DROP):

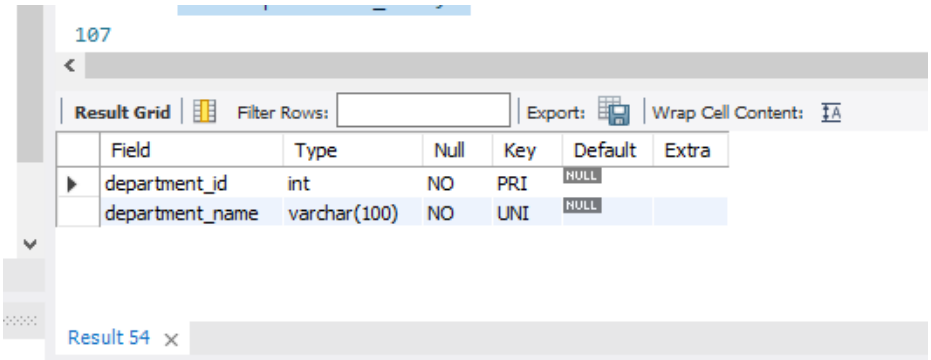
- Drop the Employees table and then the “employee” database



Constraints:

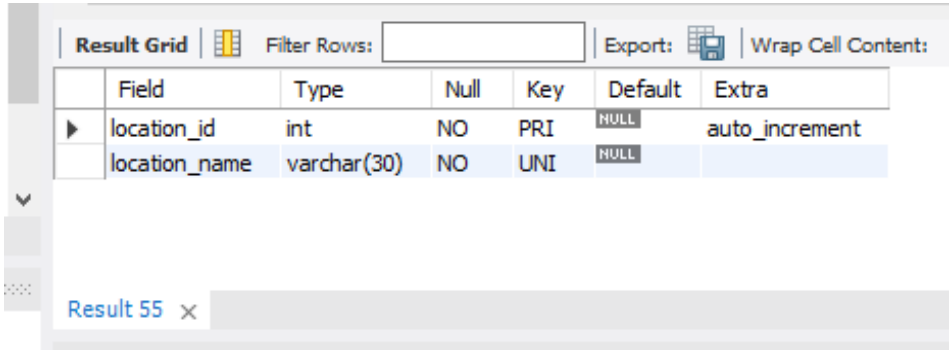
1. Departments Table:

- Ensure that the "department_id" uniquely identifies each department
- Set up constraints on the "department_name" to avoid duplicate and null entries



2.Locations Table:

- create incremented sequential for location_id
- Set up constraints on the "location_name" to avoid null and duplicate locations



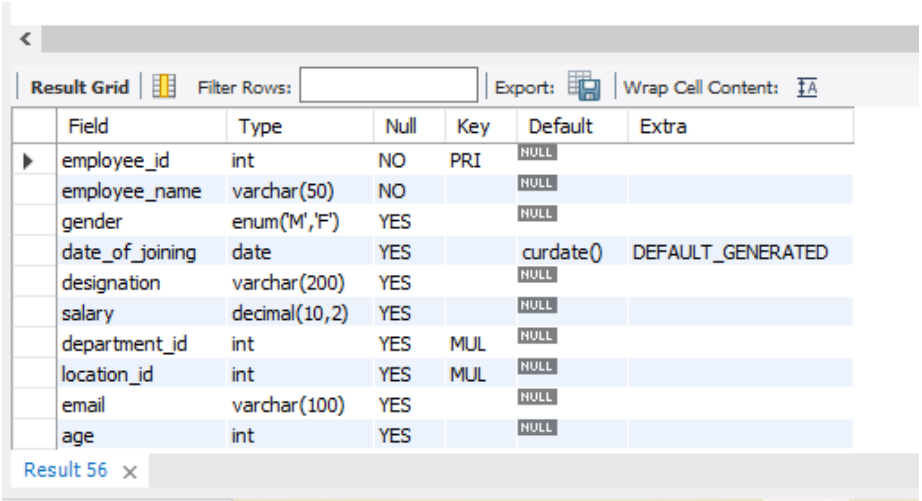
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
location_id	int	NO	PRI	NULL	auto_increment
location_name	varchar(30)	NO	UNI	NULL	

Result 55

3.Employees Table:

- Guarantee that each employee has a distinct identifier.
- Create a restriction to ensure that the employee's name is always provided.
- Limit the acceptable values for the "gender" field to only 'M' or 'F'.
- Enforce a condition to ensure that the employee's age is 18 or above.
- Automatically assign the current date to the "hire_date" field if not specified



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(50)	NO		NULL	
gender	enum('M','F')	YES		NULL	
date_of_joining	date	YES		curdate()	DEFAULT_GENERATED
designation	varchar(200)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	
email	varchar(100)	YES		NULL	
age	int	YES		NULL	

Result 56

Clause & Operators

1. DISTINCT VALUES:

● A query to retrieve distinct salaries from the Employees table

```
62
63    -- Clause & Operators:
64                                        /* 1. DISTINCT VALUES:
65                                        • A query to retrieve distinct salaries from the Employees table.*/
66
67 •  select distinct salary from employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

2. ALIAS (AS):

● Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively

Result Grid | Filter Rows: | Export: | Wra

Employee_Age	Employee_salary
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00
29	80000.00
28	60000.00
26	95000.00
33	75000.00
30	60000.00
27	85000.00
31	100000.00
28	80000.00
25	75000.00
29	55000.00
32	90000.00
26	85000.00
28	60000.00

employees 2 ×

Sorting and Grouping Data:

1. ORDER BY:

● Find employees sorted by department ID in ascending order and salary in descending order

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	date_of_joining	designation	salary	department_id	location_id	age
5020	Aarav Nair	M	2018-03-05	Software Engineer	90000.00	1	1	32
5007	Anushka Singh	F	2016-01-15	Marketing Manager	90000.00	2	3	32
5023	Anika Paul	F	2018-09-20	Public Relations Specialist	70000.00	2	2	30
5011	Saanvi Patel	F	2016-09-20	Marketing Analyst	60000.00	2	1	28
5025	Avni Iyengar	F	2019-01-20	Data Scientist	100000.00	3	4	31
5026	Vivaan Singh	M	2019-03-10	Business Analyst	75000.00	3	2	29
5001	Vihaan Singh	M	2015-01-20	Data Analyst	60000.00	3	4	27
5016	Zoya Khan	F	2017-07-05	Legal Counsel	100000.00	4	4	31
5022	Aadhya Iyengar	F	2018-07-10	HR Specialist	60000.00	4	4	28
5019	Ishika Patel	F	2018-01-15	Talent Acquisition Specialist	55000.00	4	4	29
5024	Aryan Shetty	M	2018-11-05	Product Manager	95000.00	5	1	27
5030	Sara Iyer	F	2019-11-20	Project Manager	90000.00	5	1	28
5027	Ananya Paul	F	2019-05-20	Content Writer	60000.00	6	3	32
5028	Anaya Kapoor	F	2019-07-05	Event Coordinator	60000.00	6	1	26
5015	Ishaan Kumar	M	2017-05-20	Financial Analyst	85000.00	7	1	27
5021	Advik Kapoor	M	2018-05-20	Finance Analyst	85000.00	7	3	26
5004	Kiara Malhotra	F	2015-07-05	Data Scientist	70000.00	8	3	29
5008	Diya Jha	F	2016-03-05	Graphic Designer	70000.00	8	4	27
5006	Dhruv Shetty	M	2015-11-20	UI Developer	65000.00	8	2	28
5018	Ishan Mishra	M	2017-11-20	Research Scientist	75000.00	9	3	25
5013	Arnav Rao	M	2017-01-20	Customer Success Manager	75000.00	10	3	33
5014	Vihaan Mohan	M	2017-03-10	Supply Chain Analyst	60000.00	10	2	30

employees 5

2. GROUP BY:

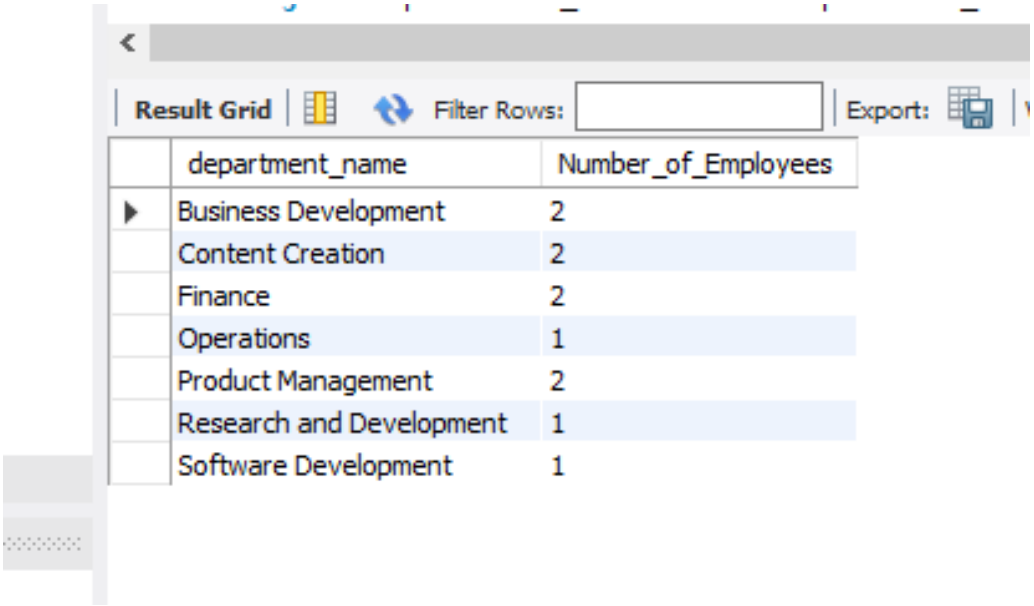
- List the maximum salary for each location

Result Grid | Filter Rows: | Export:

location_name	Maximum_Salary
Bangalore	95000.00
Chennai	95000.00
Hyderabad	90000.00
Pune	100000.00

3. HAVING:

- Find departments with less than 3 employees.



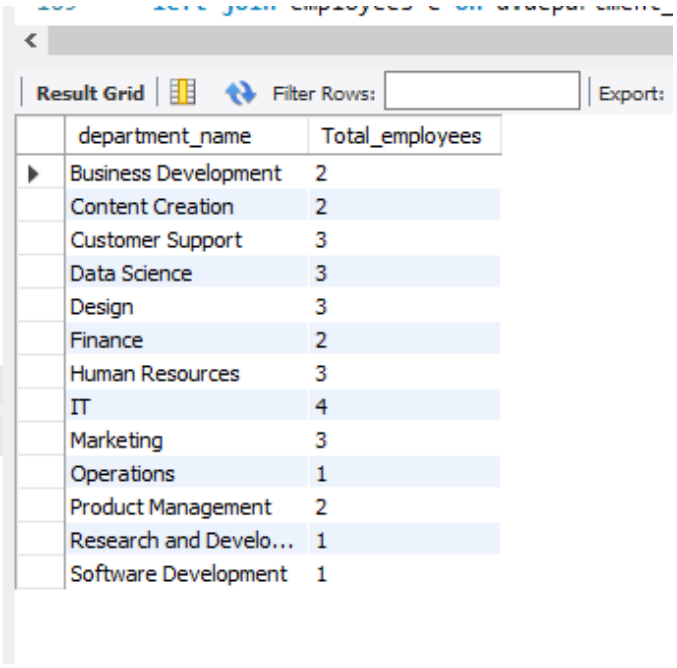
Result Grid | Filter Rows: | Export:

department_name	Number_of_Employees
Business Development	2
Content Creation	2
Finance	2
Operations	1
Product Management	2
Research and Development	1
Software Development	1

JOINS:

1. **LEFT JOIN:**

● List all departments along with the total number of employees in each department, including departments with no employees.



Result Grid | Filter Rows: | Export:

department_name	Total_employees
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3
Operations	1
Product Management	2
Research and Develo...	1
Software Development	1

2. **RIGHT JOIN:**

● Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name



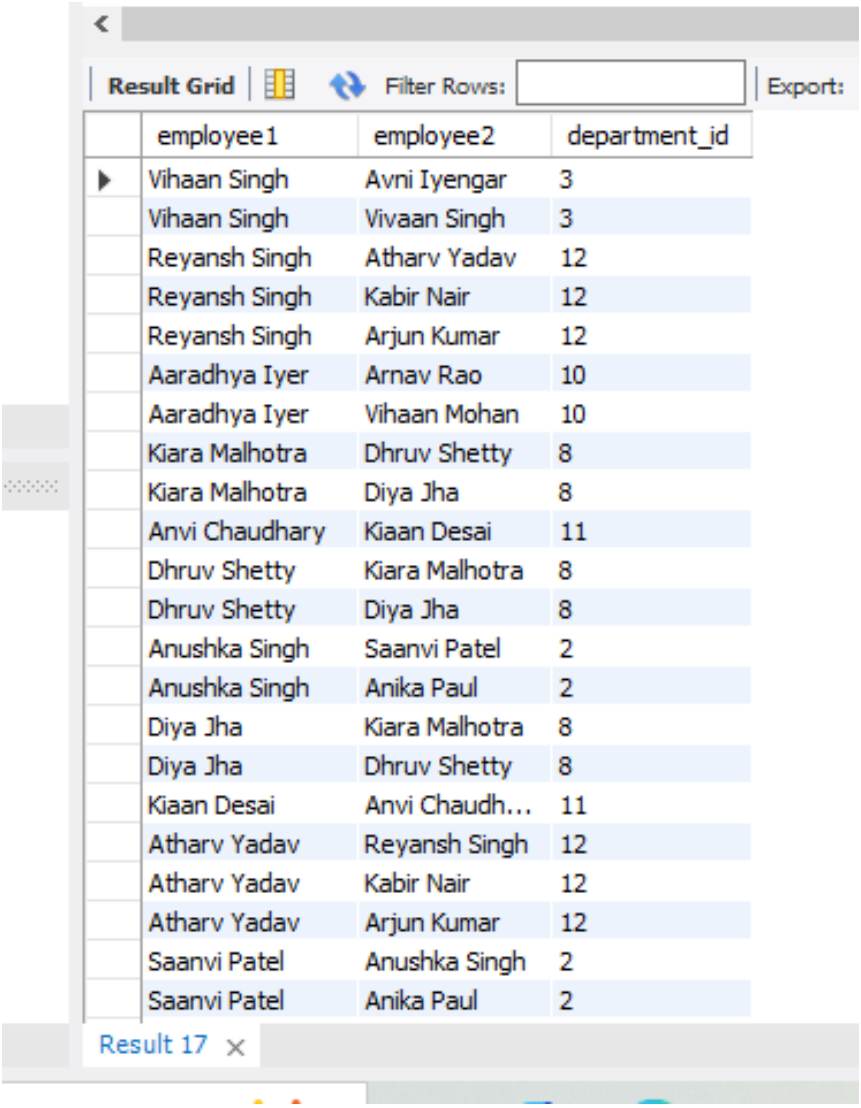
Result Grid | Filter Rows:

employee_name	location_name
Aaradhya Iyer	Bangalore
Dhruv Shetty	Bangalore
Myra Verma	Bangalore
Vihaan Mohan	Bangalore
Kabir Nair	Bangalore
Anika Paul	Bangalore
Vivaan Singh	Bangalore
Arjun Kumar	Bangalore
Reyansh Singh	Chennai
Anvi Chaudhary	Chennai
Saanvi Patel	Chennai
Ishaan Kumar	Chennai
Aarav Nair	Chennai
Aryan Shetty	Chennai
Anaya Kapoor	Chennai
Sara Iyer	Chennai
Kiara Malhotra	Hyderabad
Anushka Singh	Hyderabad
Kiaan Desai	Hyderabad
Arnav Rao	Hyderabad
Ishan Mishra	Hyderabad
Advik Kapoor	Hyderabad

Result 15

3. SELF JOIN:

- Show pairs of employees working in the same department, excluding self-pairs.



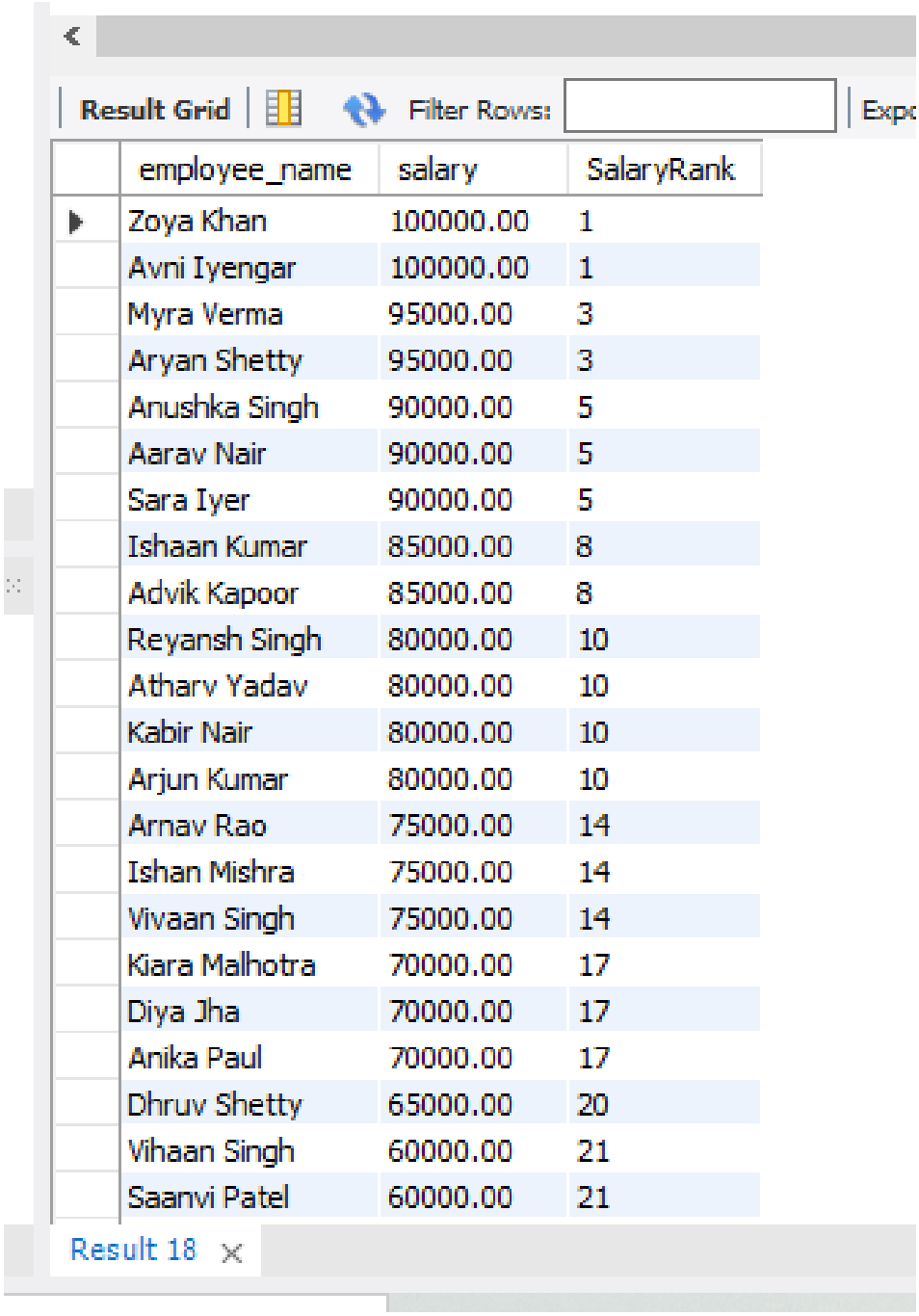
Result Grid | Filter Rows: | Export:

employee1	employee2	department_id
Vihaan Singh	Avni Iyengar	3
Vihaan Singh	Vivaan Singh	3
Reyansh Singh	Atharv Yadav	12
Reyansh Singh	Kabir Nair	12
Reyansh Singh	Arjun Kumar	12
Aaradhya Iyer	Arnav Rao	10
Aaradhya Iyer	Vihaan Mohan	10
Kiara Malhotra	Dhruv Shetty	8
Kiara Malhotra	Diya Jha	8
Anvi Chaudhary	Kiaan Desai	11
Dhruv Shetty	Kiara Malhotra	8
Dhruv Shetty	Diya Jha	8
Anushka Singh	Saanvi Patel	2
Anushka Singh	Anika Paul	2
Diya Jha	Kiara Malhotra	8
Diya Jha	Dhruv Shetty	8
Kiaan Desai	Anvi Chaudh...	11
Atharv Yadav	Reyansh Singh	12
Atharv Yadav	Kabir Nair	12
Atharv Yadav	Arjun Kumar	12
Saanvi Patel	Anushka Singh	2
Saanvi Patel	Anika Paul	2

Result 17

Windows function

- Write a window function query to rank employees by salary using rank()

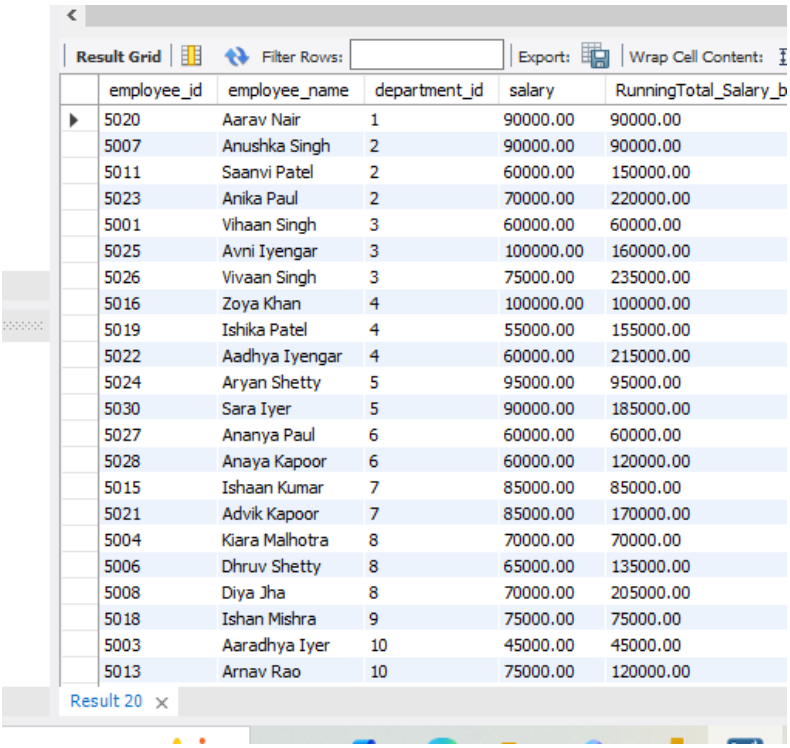


Result Grid | Filter Rows: | Expo

employee_name	salary	SalaryRank
Zoya Khan	100000.00	1
Avni Iyengar	100000.00	1
Myra Verma	95000.00	3
Aryan Shetty	95000.00	3
Anushka Singh	90000.00	5
Aarav Nair	90000.00	5
Sara Iyer	90000.00	5
Ishaan Kumar	85000.00	8
Advik Kapoor	85000.00	8
Reyansh Singh	80000.00	10
Atharv Yadav	80000.00	10
Kabir Nair	80000.00	10
Arjun Kumar	80000.00	10
Arnav Rao	75000.00	14
Ishan Mishra	75000.00	14
Vivaan Singh	75000.00	14
Kiara Malhotra	70000.00	17
Diya Jha	70000.00	17
Anika Paul	70000.00	17
Dhruv Shetty	65000.00	20
Vihaan Singh	60000.00	21
Saanvi Patel	60000.00	21

Result 18

- Write a window function query, Running total salary by department



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_id	employee_name	department_id	salary	RunningTotal_Salary_b
5020	Aarav Nair	1	90000.00	90000.00
5007	Anushka Singh	2	90000.00	90000.00
5011	Saanvi Patel	2	60000.00	150000.00
5023	Anika Paul	2	70000.00	220000.00
5001	Vihaan Singh	3	60000.00	60000.00
5025	Avni Iyengar	3	100000.00	160000.00
5026	Vivaan Singh	3	75000.00	235000.00
5016	Zoya Khan	4	100000.00	100000.00
5019	Ishika Patel	4	55000.00	155000.00
5022	Aadhya Iyengar	4	60000.00	215000.00
5024	Aryan Shetty	5	95000.00	95000.00
5030	Sara Iyer	5	90000.00	185000.00
5027	Ananya Paul	6	60000.00	60000.00
5028	Anaya Kapoor	6	60000.00	120000.00
5015	Ishaan Kumar	7	85000.00	85000.00
5021	Advik Kapoor	7	85000.00	170000.00
5004	Kiara Malhotra	8	70000.00	70000.00
5006	Dhruv Shetty	8	65000.00	135000.00
5008	Diya Jha	8	70000.00	205000.00
5018	Ishan Mishra	9	75000.00	75000.00
5003	Aaradhya Iyer	10	45000.00	45000.00
5013	Arnav Rao	10	75000.00	120000.00

Result 20

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6. Final Insights

The employee database contains **29 employees** with a total salary expenditure of **₹21,50,000**. The highest salary is **₹1,00,000**, while the minimum employee age is **25 years**. The Finance department has **2 employees** with a combined salary of **₹1,70,000**. Among the locations, **Pune** has the highest maximum salary of **₹1,00,000**, while Chennai and Bangalore each have a maximum salary of **₹95,000**. The employee distribution shows that **IT has the highest number of employees with 4**, while **Operations, Research and Development, and Software Development each have only 1 employee**. The analysis of female employees shows that **Bangalore has the lowest average female employee age at 27.33 years**, followed by Chennai at **26.75 years**, Pune at **29.20 years**, and Hyderabad at **32 years**. Four employees—Vihaan Singh (**₹60,000**), Reyansh Singh (**₹80,000**), Anvi Chaudhary (**₹55,000**), and Dhruv Shetty (**₹65,000**)—were hired before 2016 and earn more than ₹50,000. The salary and workforce distribution indicates that departments with only **1 employee**—Operations, Research and Development, and Software Development—may require attention if employee demand increases. The salary data also shows that experienced employees can have salaries ranging from **₹55,000 to ₹80,000+**, while the overall salary ceiling currently reaches **₹1,00,000**. These patterns can be used as a baseline for future workforce and salary planning. Management should review departments with **fewer than 3 employees**, particularly the three departments with only **1 employee**, and evaluate whether additional staffing is required. The **₹1,70,000 Finance salary expenditure** and the location-wise maximum salaries can be used for budget and compensation planning. Salary ranking through RANK() and department-wise DENSE_RANK() can further support identifying high-paid employees and comparing compensation within departments.

7.Conclusions

The **Employee Database Management System using MySQL** successfully demonstrates how SQL can be used to **create, maintain, validate, store, organize, and analyze employee data** effectively. The project applies database concepts such as **DDL commands,**

constraints, clauses, filtering, aggregate functions, grouping, joins, and window functions to transform employee information into meaningful business insights. The analysis covers **29 employees across 13 departments and 4 locations**, identifying important patterns in **salary, employee distribution, department-wise workforce, and location-wise employee data**. Overall, the system provides a strong foundation for **employee data management, salary and departmental analysis, workforce planning, and data-driven management decision-making**.